

Exploring Sleep Disorders: A Comparative Analysis of Machine Learning Algorithms on Sleep Health and Lifestyle Data

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Abstract— Classifying sleep disorders is crucial for improving human health, as conditions like apnea significantly impact well-being. Traditional methods are often error-prone, necessitating accurate machine learning algorithms (MLAs). This research compares several MLAs using the Sleep Health and Lifestyle Dataset. The dataset is processed by label encoding variables such as gender, occupation, blood pressure, and BMI category, followed by MinMax scaling for normalization. It is splitted into training, testing, and validation sets. K-fold cross-validation is utilized for reliable performance estimation, and hyperparameter tuning is conducted to enhance model accuracy. The parameters tuned include the number of estimators, learning rate, and maximum depth for algorithms like Random Forest, AdaBoost, and Gradient Boosting. Our findings indicate that Gradient Boosting achieved the highest classification accuracy at 93.80%, outperforming Random Forest (90.26%), Logistic Regression (91.15%), and AdaBoost (91.15%). Gradient Boosting's superior performance is attributed to its ability to sequentially build and optimize weak learners, effectively capturing complex patterns in the data. This study underscores the effectiveness of optimized MLAs in accurately classifying sleep disorders, emphasizing the importance of thorough preprocessing and hyperparameter tuning.

Keywords— BMI, blood pressure, Random Forest, AdaBoost, GradientBoost, Logistic Regression, sleeping disorder.

I. INTRODUCTION

Sleep constitutes a fundamental physiological phenomenon indispensable both for mental and physical well-being, exerting a pivotal influence in fortifying the physique and consolidating cerebral function and recollections. Nevertheless, the work efficiency of the human brain depends on the quality and duration of sleep. Sleep deprivation engenders an array of health maladies encompassing cardiovascular ailments, diabetes, and adiposity. The manual assessment of sleep stages via polysomnography (PSG) records by healthcare practitioners is labor-intensive and susceptible to fallibility, underscoring the imperative for automated methodologies.[1][2]

Philips conducts an annual survey on sleep-related attitudes and behaviors as part of World Sleep Day. In 2021, the survey included over 13,000 adults from 13 countries. The results showed that only 55% of adults were satisfied with their sleep, while the remaining 45% were dissatisfied with their sleep quality. Factors contributing to poor sleep quality included the COVID-19 pandemic, sleep apnea, and

insomnia. The survey revealed that 37% of respondents felt the pandemic negatively impacted their ability to sleep well. Additionally, 37% reported suffering from insomnia, 29% experienced snoring, 22% had a shift-work sleep disorder, and 12% dealt with sleep apnea.[3] Manual sleep stages classification entails discerning between wakefulness, light (N1), profound (N2), and profound sleep (N3) phases, as well as rapid eye movement (REM) slumber. An assortment of techniques, spanning from conventional machine learning algorithms (MLAs) to sophisticated models of deep learning, has been devised to automate sleep stage categorization.

In this study, the authors examine research on sleep disorders, highlighting challenges such as data collection. The data, often noisy and uncertain (e.g., missing data), comes from various hospitals and includes patient sleep information. However, the dataset has several limitations because it was collected from only one sleep clinic. This makes it difficult to generalize the results, as the data is biased towards certain patient groups, potentially leading to inaccurate conclusions that could affect decision-making.

The contribution of this research is to build a model on proper features that will increase the accuracy in classifying the sleep disorders.

The motivation of the research is to find the reason why people suffer the sleeping disorders most. In addition, to find the parameters that affect the sleep health of a person. Besides, to get a better way to determine sleep disorder.

II. LITERATURE REVIEW

Satapathy et al. [4] have proposed a conceptual framework for the classification of sleep stages, employing both traditional machine learning techniques and a sophisticated deep learning paradigm to automate the categorization process via the integration of multimodal signals. Within their investigation, they harnessed a repertoire of algorithms including Random Forest (RF), k-nearest Neighbors (KNN), and Support Vector Machine (SVM), alongside deep learning models combining Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architectures. This amalgamation facilitated the extraction of distinctive features from Electroencephalogram (EEG) signals via CNN, while LSTM was deployed to encapsulate the temporal intricacies inherent within the signals. Ramesh et al. [5] employed a

diverse array of machine learning algorithms (MLAs), encompassing extreme gradient boosting (XGB), light gradient boosting machine (LGBM), CatBoost (CB), Random Forest (RF), k-Nearest Neighbors (KNN), Logistic Regression (LR), and Support Vector Machine (SVM), to facilitate the early identification of individuals exhibiting elevated pretest probabilities of Obstructive Sleep Apnea (OSA), distinguishing between OSA-positive and OSA-negative cases. Utilizing the Wisconsin Sleep Cohort database as their experimental substrate, comprising a substantial cohort of 1,479 records, they meticulously scrutinized an amalgamation of clinical data, incorporating blood parameters, anthropometric measurements, and ancillary factors. Moreover, Bahrami et al. [6] introduced a novel model utilizing conventional machine learning algorithms (MLAs) including Decision Trees (DT), k-nearest Neighbors (KNN), and Random Forest (RF), alongside deep learning techniques, to detect sleep apnea utilizing data derived from single-lead Electrocardiography (ECG). Leveraging the PhysioNet ECG Sleep Apnea v1.0.0 dataset, comprising 70 individual records, the authors harnessed hybrid convolutional-recurrent Convolutional Neural Networks (CNNs) to extract salient features, while employing deep recurrent neural networks (DRNNs) to capture temporal dynamics inherent within the data sequences. Additionally, they implemented principal component analysis (PCA) to streamline the data dimensions. The hybrid architecture of CNN-DRNN exhibited superior performance in terms of accuracy detection compared to alternative algorithms explored in the study. Consequently, the authors advocated for the adoption of hybrid deep neural networks as a promising approach for the detection of sleep apnea utilizing ECG data.

In their research cited as [7], scholars utilized machine learning algorithms (MLAs) to forecast the severity of obstructive sleep apnea (OSA) syndrome utilizing proprietary data amassed from 4,014 patients, which remains undisclosed to the public. Employing a combination of supervised and unsupervised learning methodologies, including gradient boosting, Random Forest (RF), and K-means clustering, the authors achieved commendable classification accuracies of 88%, 88%, and 91%, respectively. Nonetheless, their study is encumbered by several limitations, notably the single-center origin of the data, potentially introducing biases, and the presence of missing data points. Despite these constraints, the researchers devised an MLA-based model proficient in predicting OSA severity, characterized by its efficiency and ease of use, thereby offering a promising avenue for clinical application.

Conversely, in the study referenced as [8], investigators employed a diverse array of MLAs, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, bidirectional LSTM, and gated recurrent units, to identify sleep apnea patterns utilizing data from single-lead Electrocardiography (ECG). The dataset utilized for validation purposes comprised a total of 70 records. The hybrid models proposed by the authors exhibited varying degrees of classification accuracy, with values ranging from 80.67% to 74.72%. Notably, the CNN architecture outperformed its counterparts, culminating in the identification of the hybrid CNN and LSTM network as the most effective algorithms. They postulated that routinely collected clinical metrics possess the potential to mitigate

inherent constraints. Notably, the SVM algorithm emerged as the frontrunner, achieving commendable performance metrics, including an accuracy rate of 68.06%, a sensitivity of 88.76%, a specificity of 40.74%, and an F1-score of 75.96%. Their innovative system underwent rigorous evaluation utilizing publicly available datasets such as sleep-edf. Impressively, the CNN coupled with LSTM yielded a classification accuracy of 87.4%, surpassing the performance of its algorithmic counterparts. Furthermore, the data retrieved from patient recordings exhibited inherent noise; nevertheless, the researchers implemented a Butterworth filter to ameliorate this issue, ensuring the integrity of the data for analysis.

TABLE I COMPARATIVE ACCURACY ANALYSIS

Reference No.	Algorithm	Accuracy	Dataset	Year
[4]	CNN+LS TM, RF, KNN, SVM	87.4%, 74.04%, 89.65% 76.04%	Wisconsin Sleep Cohort	2023
[6]	CNN, LSTM, MLP	68.08% 88.13%	PhysioNet ECG sleep apnea dataset	2022
[5]	SVM	68.08%	Wisconsin Sleep Cohort	2021

III. METHODOLOGY

A. Overview of the Framework

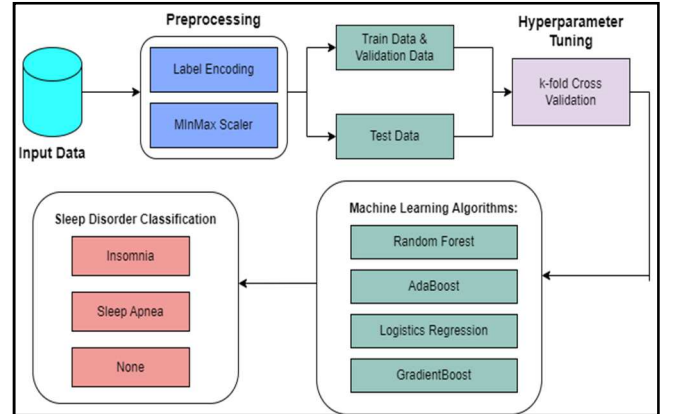


Fig. 1. Overview of the framework

In Fig. 1, first, the parameters gender, occupation, blood pressure, and BMI category from inputs are encoded using a label encoder. This approach is preferred because it efficiently transforms categorical data into numerical values, which are necessary for most machine learning algorithms. Following this, MinMax scaling is performed to normalize the feature values, ensuring that they are within a fixed range. This step

is crucial for algorithms that are sensitive to the scale of the data.

The dataset is then split into training, testing maintaining 80-20 split ratio including the validation set. Hyperparameter tuning is conducted using cross-validation to optimize model performance and avoid overfitting. Finally, various machine learning techniques, including Random Forest, AdaBoost, GradientBoost, and Logistic Regression, are applied to classify sleep disorders.

B. Hyperparameter Tuning

To optimize the model's hyperparameters and improve accuracy, we employed k-fold cross-validation. Cross-validation is a statistical method used to estimate the performance of machine learning models. In k-fold cross-validation, the dataset is divided into k equally sized folds. The model is trained k times, each time using k-1 folds for training and the remaining fold for validation.

We utilized Grid Search and Random Search methods to systematically explore combinations of hyperparameters for each algorithm, including Random Forest, AdaBoost, Gradient Boosting, and Logistic Regression. These searches were conducted within the cross-validation framework to identify the optimal hyperparameter settings that maximize accuracy.

C. Machine Learning Techniques

Random Forest: The Random Forest classifier, a robust ensemble learning method, showed comparatively lower accuracy in classifying sleep disorders in this study. This outcome is likely due to the presence of highly correlated features in the dataset, such as sleep duration, stress levels, and physical activity, which can reduce tree diversity and affect predictive power. Additionally, the modest dataset size and quality, including noise and inconsistencies, further impacted performance. Despite thorough hyperparameter tuning, the selected parameters may not have fully optimized the model for this specific dataset, indicating a need for further refinement.

AdaBoost: Adaboost exhibits remarkable versatility in its capacity to be applied to tabular data replete with both numerical and categorical features, akin to the health-related dataset presented. The performance of this MLA can be attributed to its nature of combining multiple weak classifiers to form a strong one, which is effective for reducing bias and variance. However, its sensitivity to noisy data and outliers can limit its effectiveness, as the algorithm may place too much weight on misclassified instances. Despite these challenges, AdaBoost outperformed the Random Forest model. This improvement is likely due to the effective hyperparameter tuning conducted in previous steps, which optimized the learning rate and number of estimators, enhancing the model's ability to generalize from the data.

Logistic Regression: Logistic Regression is used to classify sleep disorders in the Sleep Health and Lifestyle Dataset by modeling the probability that a given input belongs to a particular class. It works by applying a logistic function to a linear combination of the input features, which include gender, occupation, sleep duration, and stress levels. The logistic function outputs a probability value between 0 and 1, which is then thresholded to make a binary classification decision. In our study, hyperparameter tuning was performed using k-fold cross-validation to optimize the model's

performance. We specifically tuned the regularization parameter C, the solver used for optimization, and the maximum number of iterations. The optimal parameters identified were C=1, solver='lbfgs', and max_iter=200. This rigorous tuning process, combined with preprocessing steps like label encoding and MinMax scaling, helped in achieving a robust model that effectively classifies sleep disorders with high accuracy.

Gradient Boosting: Gradient Boosting is a powerful machine learning technique that sequentially builds an ensemble of weak learners, typically decision trees, to minimize prediction errors. For the Sleep Health and Lifestyle Dataset, Gradient Boosting effectively identifies patterns and relationships among various health and lifestyle factors to classify sleep disorders. We conducted hyperparameter tuning using k-fold cross-validation to optimize parameters such as the number of estimators, learning rate, and maximum depth of the trees. This process ensured the model's robustness and improved its predictive accuracy. By fine-tuning these parameters, we maximized the model's ability to generalize from the training data, leading to more reliable classification of sleep disorders.

IV. RESULT ANALYSIS

A. Dataset Description

The dataset taken for the research is the "Sleep Health and Lifestyle Dataset", consisting of 375 entries and each datum has 13 columns. Some of the features are important and some are not at all while classifying the sleeping disorder. But the gender, occupation, blood pressure, and BMI category features are taken to one hot encoding to get binary data and the standard scale is taken for a certain time.

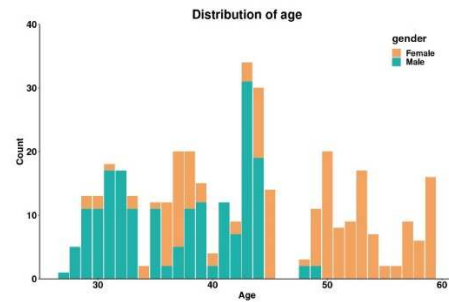


Fig. 2. Dataset distribution according to age and gender

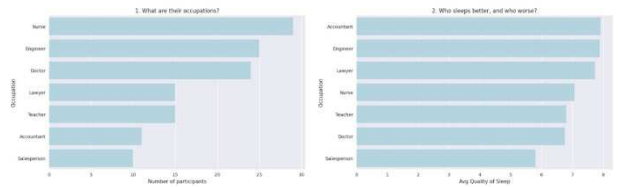


Fig. 3. Dataset distribution according to occupation

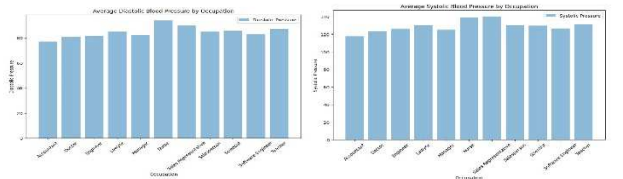


Fig. 4. Dataset distribution average systolic and diastolic pressure

Fig.2, Fig. 3, and Fig. 4 represent the dataset distribution according to age and gender, occupation, and blood pressure. In fig. 2, we can see that, most of the male are from 20 to 45 years old and most of the females are above 45. Fig. 3 shows that accountants, engineers, and lawyers have the best sleeping experience and salespersons have the worst. Fig. 4 shows the average systolic and diastolic pressure for each occupation.

B. Correlation Matrix

The correlation matrix shown in Fig. 2 elucidates the intensity and directionality of associations among diverse variables. Specifically, noteworthy positive correlations manifest between age and both sleep duration (0.34) and sleep quality (0.47), as well as between sleep duration and quality (0.88), and physical activity level and daily steps (0.77).

These findings imply that advancing age correlates with prolonged sleep duration and enhanced sleep quality, alongside heightened physical activity being linked to increased daily step counts. Conversely, conspicuous negative correlations emerge between stress level and both sleep duration (-0.81) and sleep quality (-0.90), as well as between stress level and heart rate (-0.67). Such negative correlations indicate that heightened stress levels coincide with diminished sleep duration, inferior sleep quality, and reduced heart rates. In essence, the correlation matrix furnishes insights into the intricate interplay among myriad factors influencing sleep, physical activity, and stress levels within the dataset.



Fig. 5. Correlation matrix

C. Evaluation of Performace

From Fig. 6, it is observed that the feature that directly affects the quality of sleep of a person is the blood pressure. Normal systolic and diastolic pressure ensure a better sleeping experience. The second most important feature is the BMI category. A better BMI indicates sound sleep and an abnormal BMI index usually causes sleeping disorder. Occupation, age, and physical activity level also influence the duration of sleep. Occupation having more physical activity causes more tiredness and causes sleep. Infants and older people sleep more than normal sleeping periods. Gender is

the least important factor in the sleeping disorder. People of the same age and same gender sleeps almost the same duration.

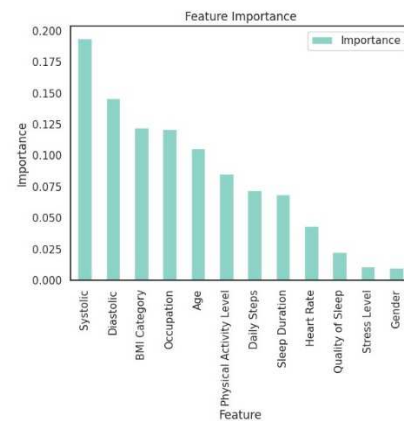


Fig. 6. Importance of features based on the correlation matrix

Random Forest: The code performs hyperparameter

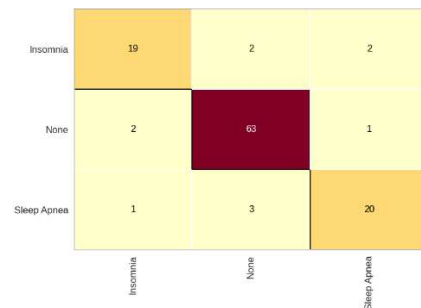


Fig. 7. Confusion matrix (Random forest)

tuning for a Random Forest classifier using Randomized Search Cross-Validation, exploring various combinations of maximum depth, minimum samples split, and criterion. It identifies the best-performing parameters and prints the selected criterion, the corresponding accuracy score, and the optimal values for minimum sample split and maximum depth. The accuracy is 90.26% and the confusion matrix is shown in Fig. 7.

AdaBoost: The code initializes an AdaBoost classifier with 500 weak learners, a learning rate of 0.02, and sets the random state to 0 for reproducibility. Accuracy is 91.15% and the confusion matrix is projected in Fig. 8.

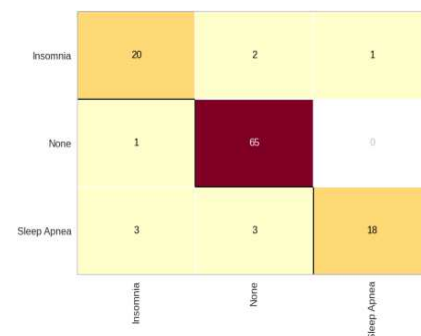


Fig. 8. Confusion matrix (Adaboost)

Gradient Boosting: The code initializes a Gradient Boosting classifier with 300 weak learners, a learning rate of 0.01, and sets the random state to 0 for reproducibility. The

accuracy that we have found is 93.80% and this accuracy outnumbers all the other accuracies.

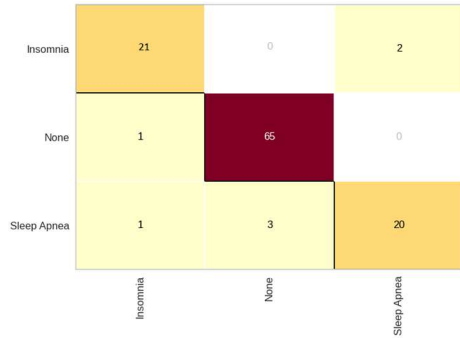


Fig. 9. Confusion matrix (Gradient Boosting)

In classification tasks, Gradient Boosting often outperforms Random Forest, Logistic Regression, and Adaboost due to its ability to sequentially improve model predictions by focusing on the instances that are difficult to classify. The classification report for these MLAs are given below:

TABLE II. COMPARISON OF ACCURACY

Approach	Accuracy	Approach	Accuracy
Random Forest	90.26%	AdaBoost	91.15%
Logistic Regression	91.15%	GradientBoost	93.80%

TABLE III. CLASSIFICATION REPORT

Algorithm	Precision	Recall	F1-score
Random Forest	0.91	0.89	0.90
AdaBoost	0.92	0.91	0.91
Logistic Regression	0.92	0.91	0.91
GradientBoost	0.93	0.92	0.92

TABLE IV. COMPARISON OF ACCURACY ON THE SAME DATASET

Author(s)	Dataset	Method	Accuracy
Talal et al. [9]	Sleep Health and Lifestyle	ANN	92.92%
Hidayat [10]	Sleep Health and Lifestyle	Random Forest	88.3%
Our Study	Sleep Health and Lifestyle	K-fold cross validation + Gradient Boosting	93.80%

V. CONCLUSION

In conclusion, our research on the Sleep Health and Lifestyle dataset revealed intricate relationships between various factors influencing overall well-being.

Through comprehensive analysis, we identified age as a significant predictor of both sleep duration and quality, alongside other parameters such as blood pressure, BMI, and gender. The research faced challenges in collecting data with appropriate features, avoiding misleading data, and selecting suitable encoding techniques. For preprocessing, label encoding was applied to categorical variables, and MinMax scaling was performed to normalize the data. The classification accuracy scores achieved by the respective algorithms were as follows: Random Forest (90.26%), Logistic Regression (91.15%), AdaBoost (91.15%), and Gradient Boosting (93.80%). Gradient Boosting achieved the highest accuracy among the evaluated algorithms, demonstrating superior performance overall. Cross-validation techniques, particularly k-fold cross-validation, were employed to tune hyperparameters and enhance model robustness, ensuring reliable and generalized performance. For future work, we recommend using a larger dataset to improve model accuracy and exploring the application of neural networks for building more effective models.

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